

Conservative Dentistry & Endodontics

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Departmental Objective:

1. To teach Clinical topics alongside theory in order to integrate between theory and practice.
2. Introduced of the clinical students to patient care in a carefully controlled environment
3. To learn about and carry out more advanced procedures in restorative dentistry and Endodontics
 4. Final objective of this curriculum is to teach current and new concepts of
 - i. Operative and endodontic care
 - ii. Aesthetic dentistry
 - iii. Dental radiology

By training dental students to be able to practice four-handed dentistry effectively and efficiently.

List of Competencies to acquire:

This curriculum is designed to acquire following competencies:

1. Communicate properly to obtain a medical and psychosocial history, and will incorporate this information to frame an effective treatment plan for individual patient.
2. Demonstrate diagnostic skills.
3. Provide emergency care and control pain.
4. Demonstrate restorative skill.
5. Demonstrate endodontic skill.
6. Manage patient under conscious sedation.
7. Perform Manual skill and dexterity.
8. Manage patients with complex dental problems by referral or by working under supervision of appropriate dental instructor.
9. Practice Minimally Invasive Dentistry (MID) & Evidence Based Dentistry (EBD) with scientific understanding and thought.
10. Work in a team and shows professionalism.

Distribution of teaching - learning hours

Lecture	Tutorial	Practical	Clinical Teaching	Total Teaching hours	Integrated teaching (Common)	Assessment exams (Formative)		Professional exam (Summative)	
						Preparatory leave	Exam time	Preparatory leave	Exam time
90 hrs	50 hrs	28 hrs	96 hrs	264 hrs	10 hrs	10 days	20 days	10 days	35 days

Teaching - learning methods, teaching aids and evaluation

Teaching Methods			Teaching aids	In course evaluation
Large group	Small group	Self learning		
Lectures	Tutorial, practical and clinical classes	Assignment- based self-directed learning	• Text books • Models • Simulators • Patients • Journals • Online library	• Item • Card final • Ward final • Mid-term assessment examination • Final assessment examination

Related Equipment:

1. Multimedia
2. Over Head Projector (OHP)
3. Slide Projector
4. Video
5. Microphone with speaker
6. Digital Board
7. White Board
8. Black Board
9. X-Ray Plate with View Box
10. Specimen etc.

Marks Distribution for Final Professional Examinations of Conservative Dentistry & Endodontics:

Written 100 marks	Formative Assessment	Mid term exam	4 marks
		Final term exam	4 marks
		Attendance	2 marks
	MCQ		20 marks
	SAQ		70 marks
SOE 100 marks	Board I		50 Marks
	Board II		50 Marks
Practical and Clinical 100 marks	OSCE		40 marks
	Log Book		5 marks
	Assignment		5 marks
	Long Case		50 marks

Learning Objectives and Course Contents in Conservative Dentistry and Endodontics Endodontics

Learning Objectives	Contents	Teaching Hours
The students should be able to – - Define endodontium, periodontium and endodontics. - Explain the scopes of endodontics. - Recognize the recent advancements in instruments, materials and diagnostic procedures of endodontics.	1. (a) Introduction and scopes of Endodontics - Endodontium, Periodontium and Endodontics. - Scopes of endodontics: - Protection of pulp and periradicular tissue. - Diagnosis and management of dental pain. - Management of dentine hypersensitivity. - Tooth infractions - Direct and indirect pulp treatments. - Restoration of endodontically treated tooth. - Management of trauma to tooth and associated structures. - Surgical Endodontics. - Bleaching - Endodontic management of tooth resorption	L - 1.5 hrs
	1. (b) Modern Endodontics - Instruments - Materials - Diagnostic aids - Laser in Dentistry	T - 1.5 hrs
The student should be able to – - Take history of a patient. - Perform examinations needed to diagnose a case. - Evaluate a case prior to endodontic treatment. - Describe the diagnostic aids and identify the suitable method to diagnose a case.	2. Patient assessment and diagnostic procedures for endodontic therapy - Considerations prior to endodontic therapy - History - Clinical examination - Investigations	L - 3.0 hrs T - 1.5 hrs P +C = 2+3 hrs

Learning Objectives	Contents	Teaching Hours
The students will be able to – • Outline the sequelae of dental caries • Categorize pulpal and periapical pathosis • Describe and distinguish the clinical and radiological features of different pulpal and periapical pathosis. • Diagnose pulpal and periradicular pathology. • Treat the different pathosis. • Describe the communications and inter-relationships of endodontium and periodontium • Mention the causes, contributing factors, classification, differential diagnosis, management and prognosis of different endo-perio and perio-endo lesions.	3.(a) Pulpal and periradicular pathosis • Introduction • Reversible pulpitis • Irreversible pulpitis • Hyperplastic pulpitis • Apical periodontitis • Periapical abscess • Periapical granuloma • Radicular cyst 3.(b) Endodontic-Periodontal interrelationships • Endodontic-periodontal Communications • Pulpal-periodontal interrelationships, • Periodontal-pulpal interrelationships • Aetiological factors • Contributing factors • Classification • Differential Diagnosis • Management • Prognosis	L - 3.0 hrs T –1.0 hrs
The students should be able to – • Describe morphology and pulp space anatomy of different teeth. • Illustrate the types of canal configuration and apical anatomy. • Describe the common anomalies of the pulp cavity and root canal system.	4. Pulp space anatomy • Morphology of teeth and their root canal system • Types of canal configuration • Apical anatomy • Anomalies of pulp cavities	L - 1.5 hrs T - 1.5 hrs

Learning Objectives	Contents	Teaching Hours
The student should be able to - • Define RCT. • List the objectives and steps of RCT • Describe the indications and contraindications of RCT • Mention the purpose of single visit RCT. • Describe the initial procedures needed to perform before an endodontic treatment	5.(a) Definition, objectives, indications and contraindications of RCT Definition, objectives, steps, indications and contraindications of RCT 5.(b) Single visit RCT Advantages, disadvantages, indications and contraindications. 5.(c) Pre-endodontic preparations	L - 1.5 hrs T - 1.5 hrs
Students should be able to – • Enlist and classify the endodontic instruments • Identify different endodontic instruments and mention their use. • Mention different types of irrigating solutions with their features • Describe the purpose of irrigation and properties of irrigating solutions • Identify ideal irrigant • Describe the needles and proper methods used to irrigate • Recognize modern irrigating solutions • Mention the types of different chelating agents • Discuss the properties and role of chelating agents • Identify ideal chelating agents • Recognize modern chelating agents • Mention the name of different lubricating agents • Describe the purpose of lubrication • Mention the role and different types of intracanal medicaments • Describe the features of common intracanal medicaments • Identify the ideal intracanal medicament • Recognize recent intracanal medicaments	6.(a) Endodontic instruments and Armamentarium • Endodontic kit • Burs • ISO specified instruments 6.(b) Irrigation Irrigation and irrigating solutions 6.(c) Chelation Chelation and chelating agents 6.(d) Lubrication and intracanal medication • Canal lubrication and lubricating agents • Intracanal medication and medicaments	L - 3.0 hrs T - 1.5 hrs P+C= 2 +8hrs

Learning Objectives	Contents	Teaching Hours
Students should be able to – • Describe the nerve supply of different teeth. • Describe the types of LA and its composition, mechanism and mode of administration. • Describe the roles of its components and the complications of LA • Describe the routine approach to conventional local anaesthesia: when and how to anaesthetize. • List techniques that are helpful in giving painless injections. • Mention how to obtain anesthesia for different endodontic procedures.	7. Local anaesthesia related to operative dentistry and endodontics • Nerve supply of teeth • Classification of LA • Composition • Mechanism of action • Techniques • Failure of analgesia • Complications • Mode of administration	L - 2.0 hrs T-1.5 hrs P+C = 2+8hrs
Students should be able to – • Mention the basic principles of an access cavity preparation. • Describe and illustrate the access cavity preparation techniques of different teeth and the possible errors during preparation.	8. Access cavity preparation • Principles of access cavity preparation • Access cavity preparation of different teeth • Errors	L – 3.0 hrs P+C = 2+8hrs
Students should be able to – • Define CRP, ASI, major diameter, minor diameter, radiographic apex, anatomic apex. • Describe the significance of working length determination. • List the different working length determination procedures. • Describe and determine the working length via the common techniques.	9. Working length determination • CRP • ASI • Significance • Methods	L - 1.5 hrs P+C =2 +8hrs

Learning Objectives	Contents	Teaching Hours
Students should be able to – • Define and describe biomechanical (chemomechanical) preparation of root canal and recapitulation. • Describe smear layer produced during canal preparation. • Describe different types of root canal preparation techniques. • Solve problem regarding canal preparation in abnormal situations.	10. Preparation of Root Canal System • Chemomechanical preparation • Canal preparation through instrumentation	L - 3.0 hrs T - 1.5 hrs P+C =2 +10hrs
Students should be able to – • Recognize the clinical criteria that determine when to obturate. • Define root canal obturation, overextension and overfilling. • Describe the ideal criteria of obturating materials and purpose of obturation. • Classify the different obturating materials and techniques. • Describe the techniques of mixing and placing a sealer. • Describe and differentiate between lateral and vertical condensation technique and suggest where each is indicated. • Describe the clinical and radiographic criteria for evaluating the quality of obturation.	11. Obturating materials and root canal obturation • Root canal obturation • Readiness for obturation • Sealing and sealers • Filling and fillers • Obturating materials and techniques	L - 1.5 hrs T - 1.5 hrs P+C = 2 +10hrs
Students should be able to – • Define the terms and virulent factor associated with endodontic microbiology • Describe the portals of entry of microorganisms into pulp and periradicular tissues • Describe the microfloras of root canal and microorganisms associated with for endodontic infection. • Classify the commonly used antibiotics, analgesics and antiulcerants with their mechanism of action, dose and duration. • Select the appropriate antibiotics, analgesics and antiulcerants and describe their mechanism, advantages, adverse effects and contraindications. • Describe the purpose and indications of antibiotic prophylaxis. • Name the antibiotics used in antibiotic prophylaxis. • Make adjunctive therapeutic selections when required and write a prescription.	12. Microbiological and pharmacological aspects of endodontics • Root canal microflora • Portal of entry of microorganisms • Indication of antibiotics in endodontic procedures • Antibiotics, analgesics and anti-ulcerants • Pregnancy and drugs • Antibiotic prophylaxis	L - 1.5 hrs T - 1.5 hrs

Learning Objectives	Contents	Teaching Hours
Students should be able to – • Define endodontic mishap. • Describe different types of endodontic mishaps and their management. • Evaluate and estimate prognosis before during and after different endodontic treatment. • Describe the criteria of success or failure of an endodontic treatment.	13. (a) Endodontic Mishaps, • Definition • Types • Management of mishaps 13.(b) Endodontic Retreatment	L - 2.0 hrs T - 1.5 hrs
Students should be able to – • Define and describe pulp capping, pulpotomy, apexogenesis and apexification. • Describe and differentiate between open and closed apices. • Describe methods of diagnosis and selection of appropriate treatment option. • Recognize the success and failure of treatment of an open apex. • Justify the necessity of root canal obturation after performing apexogenesis or apexification	14.Pulp capping, Pulpotomy, Apexogenesis, Apexification • Definition • Indications • Procedure	L - 2.0 hrs T - 1.5 hrs
Students should be able to – • Classify and describe the clinical and radiographic features and management of various types of tooth injury. • Describe the differences in history taking for different injuries. • Describe the diagnostic tests and procedures used in examining different traumatic injuries of tooth.	15.(a) Traumatic Injury of tooth • Classification • Management	L - 3.0 hrs T - 1.5 hrs
Students should be able to – • Define and describe the emergencies that require endodontic approach. • Identify the causes of emergencies as they occur before, during and after endodontic treatment. • Describe the immediate treatment of these emergencies. • Outline the treatment options and supportive therapy for these conditions.	16. Endodontic Emergency • Definition • Reversible pulpitis • Acute irreversible pulpitis • Acute periapical periodontitis • Acute periapical abscess • Tooth infractions • Avulsion • Intra treatment pain, Endodontic Flare up	L -2.0 hrs

Learning Objectives	Contents	Teaching Hours
Students should be able to – - List the different types of restorative options to restore endodontically treated teeth. - Outline post-operative risks of unrestored teeth. - Describe the role of restoration in longevity of endodontically treated teeth. - Describe requirements of an adequate restoration and how it protects and seals coronally.	17. Restoration of Endodontically Treated Tooth - Intracoronar restorations - Extracoronar restorations	L - 3.0 hrs T - 1.5 hrs
Students should be able to – - Describe the roles of endodontic surgery as compared to non-surgical root canal therapy. - Define the terms listed in the content. - Classify and describe each procedure listed. - List the commonly used root-end filling materials.	18. Endodontic surgery - Indications - Contraindications - Classifications - Principles of flap design - Classification of surgical flaps - Apisectomy/ Root end resection - Retrograde obturation - Hemisection, Bisection & Radisectomy - Reimplantation - Intentional reimplantation - Transplantation & Trephination	L - 3.5 hrs T- 1.5 hrs
Students should be able to – - Define, classify and manage tooth resorption - List the etiology of resorption.	19. Endodontic management of Tooth Resorption - Definition - Types - Management	L - 1.5 hrs T - 1.5 hrs
Students should be able to – - Describe the various causes and mechanism of tooth discoloration with their management. - Describe means of preventing coronal discoloration. - Describe the bleaching agents with their application procedures and complications. - Evaluate the prognosis of bleaching treatments.	20. Discoloration of teeth and its management - Definition - Cause - Bleaching - Bleaching agents - Mechanism of bleaching - Indication and procedures of different bleaching	L - 2.0 hrs T - 1.5 hrs

OPERATIVE DENTISTRY

Learning Objectives	Contents	Teaching Hours
Students should be able to – - Define operative dentistry. - Describe indications, pre-operative considerations and future of operative dentistry. - List the commonly used restorative materials in operative dentistry. - Define the direct, indirect, intracoronal, extracoronal, permanent, intermediate and temporary restorations. - Define Minimally Invasive Dentistry (MID), - Describe Nano-Dentistry, Evidence Based dentistry (EBD)	1. Introduction and scopes of operative dentistry - Operative Dentistry - Need for operative dentistry - Considerations prior to operative treatment - Future of operative dentistry - Restorative Dentistry - Direct & indirect - Intracoronal & extracoronal - Permanent, intermediate and temporary restorations - Minimally Invasive Dentistry (MID) & Evidence Based Dentistry (EBD)	L - 2.0 hrs T - 2.0 hrs
Students should be able to – - Mention the criteria of patient selection and assessment. - Describe the different diagnostic aids related to operative dentistry. - Outline the treatment modalities for different conditions.	2. Patient Assessment and Diagnostic Procedures for operative dentistry, Treatment Planning - Considerations related to patient selection for treatment - History - Clinical examination - Investigation for diagnosis	L - 2.0 hrs T - 2.0 hrs C - 2 hrs
Students should be able to – - Explain the necessity of restoring the occlusal and periodontal anatomy. - Define the terms listed in the content. - Describe the traits and importance of each.	3. Occlusal and periodontal aspects of restorative dentistry - Centric relation, centric occlusion, normal occlusion. - Biologic width, tooth contour, contact point, embrasure, marginal ridge, restoration margins, surface finishing.	L - 1.5 hrs T - 1.5 hrs

Learning Objectives	Contents	Teaching Hours
Students should be able to – • Define and classify dental caries. • Describe the aetiologic factors of dental caries. • Describe the histopathology of enamel and dentine caries. • Differentiate between infected and affected dentine. • Describe the different types of dental caries. • Describe the diagnostic measures to identify carious lesions. • Describe the treatment options and preventive measures for dental caries. • Illustrate the sequelae of dental caries.	4. Dental Caries • Definition • Etiology • Pathophysiology • Classification • Histopathology • Diagnosis • Prevention • Treatment • Sequelae	L - 2.0 hrs T - 2.0 hrs
Students should be able to – • Determine the best position for both the patient and the operator during operative treatment. • Control pain and describe the methods available for controlling pain. • Describes the purpose of isolation. • Describe the fields that need to be isolated. • Describe the various methods of isolation. • Describe rubber dam in details. • Describe the ideal criteria of matrix band. • Classify and describe the types, roles and indications of different types of matrix bands, retainers and wedges. • Identify the operative instruments and describe their uses.	5. Preliminary considerations for operative dentistry • Patient and operator position • Pain control • Isolation of the operating field • Operative instruments	L - 4.0 hrs T - 4.0 hrs P+C =2 + 3hrs

Learning Objectives	Contents	Teaching Hours
Students should be able to – • Define the tooth preparation terminologies. • Classify and illustrate cavities. • Describe the objectives and factors of tooth preparation. • List, define and describe the steps of tooth preparation. • Define different types of tooth preparation.	6. Tooth preparation • Definition • Objectives • Factors affecting • Tooth preparation terminology • Classification of tooth preparation • Stages of tooth preparation(Principles of cavity preparation)	L - 4.0 hrs T - 4.0 hrs
Students should be able to – • Define class I and II cavities. • Describe the armamentarium needed to prepare the cavities. • Describe the pre-operative procedures. • Describe and illustrate the preparation procedure according to the principles of cavity preparation. • Prepare class I and II cavities.	7. Class I and Class II cavities • Definitions • Armamentarium required • Initial clinical procedures • Preparation Procedures	L - 1.5 hrs P +C= 2 hrs
Students should be able to – • Define alloy, amalgam and dental amalgam. • Describe the composition and setting reaction of silver amalgam along with the role of the components. • Describe the types, uses and benefits of different types of silver alloy. • Describe the steps of restoration with silver amalgam. • Describe the hazards and maintenance of mercury.	8. Silver amalgam and silver amalgam restoration procedure • Alloy, amalgam, dental amalgam. • Composition and setting reaction of silver amalgam • Types of silver amalgam • Steps of restoration	L - 1.5 hrs P+C =2 + 6hrs
Students should be able to – • Differentiate between sterilization and disinfection. • Describe the common methods of sterilization in conservative dentistry. • Differentiate between these methods to determine which one is better. • Describe the different sterilization methods for different operative and endodontic instruments. • List the possible transmissible diseases. • Describe the preventive protocol against cross infection.	9. Sterilization and prevention of cross infection • Definition of sterilization, disinfection and cross-infection. • Methods of sterilization • Sterilization of operative and endodontic instruments. • Transmissible diseases • Prevention of cross infection	L-3.0 hrs P+C =2 hrs

Learning Objectives	Contents	Teaching Hours
Students should be able to – • Define x-ray, radiograph, ionizing radiation and dosimetry. • Describe the properties of x-ray • Describe the effects of radiation on biologic tissues. • Describe the radiation prone tissues and hazards of radiation. • Describe the preventive measures against radiation. • Describe the types of dosimetry. • Describe radiation hazards and protective measures	10. Radiology related to operative dentistry and endodontics – I • Introduction to dental radiology & X-ray production properties of X-rays. • Definitions • Effects of radiation on tissues • Hazards of radiation • Dosimetry • Radiation protection • Recent advancement in Dental Radiology	L- 2.0 hrs T - 2.0 hrs
Students should be able to – • Define class III and IV cavities. • Describe the armamentarium needed to prepare the cavities. • Describe the pre-operative procedures. • Describe and illustrate the preparation procedure according to the principles of cavity preparation. • Prepare class III and IV cavities. Students should be able to – • Mention the indications of pin retained restoration. • Describe the types of pins for restoration. • Describe the procedure of pin setting. • Describe dentatus screw.	11.(a) Class III and Class IV cavities • Definitions • Armamentarium required • Initial clinical procedures • Preparation Procedures 11.(b) Pins • Indications • Types and procedure	L -2.5 hrs P+C =2+4hrs
Students should be able to – • Define class V and VI cavities. • Describe the armamentarium needed to prepare the cavities. • Describe the pre-operative procedures. • Describe and illustrate the preparation procedure according to the principles of cavity preparation. • Prepare class V and VI cavities.	12. Class V and Class VI cavities • Definitions • Armamentarium required • Initial clinical procedure • Preparation Procedure	L-1.5 hrs P+C =2+4hrs

Learning Objectives	Contents	Teaching Hours
Students should be able to describe – - Minimally Invasive Dentistry (MID), - Nano-Dentistry, Evidence Based dentistry (EBD) - Lip Line - Smile Design - Face profile - Colour of Gingiva - Translucency Of the tooth - Shade selection - Aesthetics, Health & Function of the Tooth	13. Aesthetic Dentistry - Color - Translucency - Aesthetics and operative dentistry-enameloplasty, bleaching restoration with composite resin, veneers, full coverage crown	L - 1.5 hrs T - 1.5 hrs
Students should be able to – - Describe the roles of adhesive dentistry. - Describe the agents used for adhesion with their roles in enamel and dentine adhesion. - Differentiate between the enamel and dentine adhesion. - Describe hybrid layer and smear layer. - Recent advancements of enamel and dentine adhesion. - Mention the causes of debonding along with prevention. - Describe the properties, composition, reactive agents, types and uses of composite resin. - Describe the restoration procedure. - Describe hybrid composite. - Describe recent advancements.	14. (a) Enamel and Dentine Adhesion - Scopes of adhesive dentistry - Agents used for adhesion - Steps of enamel and dentine adhesion Debonding 14. (b) Composite resins and their restoration procedures - Properties - Composition, reactive components, types, uses - Restorative procedure	L - 3.0 hrs P+C =2+4hrs
Students should be able to – - Describe the composition and setting reaction of GIC. - List the different types of GIC with their uses. - Describe the restorative procedure. - Describe ART and RMGI. - Mention role of GI conditioner, GI varnish or petroleum jelly. - Describe sandwich technique and determine when it is necessary.	15. Glass ionomer cement and its uses - Composition - Types and uses - Setting reaction - Restorative procedure - ART - Sandwich technique	L - 2.0 hrs P+C =2+4hrs

Learning Objectives	Contents	Teaching Hours
Students should be able to – • Define microleakage. • Describe its clinical effects and causes. • Describe the measures of prevention of microleakage in Ag-amalgam, composite and GIC.	16. Microleakage • Definition, causes, clinical significance of microleakage • Prevention of microleakage in restorations.	L - 1.5 hrs
Students should be able to – • Define different types of non-carious tooth surface lesions. • Determine the factors responsible for the lesions. • Learn the measures for taking history and diagnosis. • Describe the findings, prevention and treatment options of the lesions.	17. Non-carious tooth surface loss • Definitions, Aetiology, Clinical features, Prevention and Treatment of attrition, abrasion, erosion and abfraction.	L-2.0 hrs T-2.0 hrs
Students should be able to – • Define dentine hypersensitivity. • Describe the prevalence and causes of dentine hypersensitivity. • Describe the dentine sensitivity theories. • Mention prevention and management options.	18. Dentine Hypersensitivity and its management • Definition • Aetiology • Clinical features • Management	L- 1.5 hrs
Students should be able to – • Describe anatomic and functional relationship between dentine and pulp. • Determine which factors may cause pulp injury during operative procedure and after restoration. • Describe the preventive measures to protect pulp during operative procedure and after restoration. Students should be able to – • Describe the effects of the mentioned restorative materials on pulp. • Define liners, bases and cavity varnish. • Describe different types of liners, bases, cavity varnish and tubule blocking agents with their roles in protecting pulp. • Determine which lining or base or cavity varnish is ideal to use in case of different restorative materials.	19.(a) Pulp protection • Prevention of pulp injury during cavity preparation • Prevention of pulp injury during and after restoration. 19. (b) Pulpal response to restorative materials Pulpal response to Ag-amalgam, composite, GI, Liners, Bases and Cavity Varnish, Tubule blocking agents.	L -2.0 hrs

Learning Objectives	Contents	Teaching Hours
Students should be able to – • List the various intra-oral and extra-oral radiographic techniques. • Enumerate the features seen in a periapical view, bitewing x-ray and OPG. • Describe the indications, advantages and disadvantages of periapical view, bitewing x-ray, OPG and occlusal view. • Differentiate between bisecting and paralleling radiographic technique. • Describe the tube shift principle. • Describe different radioopaque and radiolucent structures. • Identify X-ray films & position of the teeth • Mention the steps of processing an x-ray film. • Describe and identify the radiological features of different pulpal and periradicular lesions. • Identify defects, faults and artifacts of various x-ray films	20. Radiology related to operative dentistry and endodontics - II • General technical considerations. • Extra oral & Intra oral dental radiology, OPG, • Normal radioopaque and radiolucent structures • Interpretations of differential diagnosis of different X-ray • Identifying of X-ray films • Processing of X-Ray Films • Radiographic defects, faults and artifacts	L - 4.0 hrs T - 4.0 hrs P+C =2+10hrs